

6.4: Discrete Conditional Distributions

Remember, for any two events E, F with $P(F) > 0$ we say the conditional probability of E given

$$\text{is } P(E|F) = \frac{P(E \cap F)}{P(F)}$$

For discrete random variables X and Y we can naturally define a conditional probability distribution: $p_{X|Y}(x|y) = P(X = x|Y = y) = \frac{P(X = x, Y = y)}{P(Y = y)}$

Notice that if X, Y are independent, then $P(X = x, Y = y) = P(X = x)P(Y = y)$, and thus $p_{X|Y}(x|y) = p_X(x)$

Example: Consider two random variables $X, Y \in \{0, 1\}$ with joint probability mass function

$$p(0, 0) = \frac{1}{3}, p(0, 1) = \frac{1}{6}, p(1, 1) = \frac{1}{12}, p(1, 0) = \frac{5}{12}$$

Evaluate each of $p_{X|Y}(a|b)$ and $p_{Y|X}(a|b)$ for $a, b \in \{0, 1\}$. Are X and Y independent?

	$x=0$	$x=1$
$y=0$	$1/3$	$5/12$
$y=1$	$1/6$	$1/12$

$$P(Y=0) = \frac{1}{3} + \frac{5}{12} = \frac{3}{4}$$

$$P(Y=1) = \frac{1}{6} + \frac{1}{12} = \frac{1}{4}$$

$$P(X=0) = \frac{1}{3} + \frac{1}{6} = \frac{1}{2}$$

$$P(X=1) = \frac{5}{12} + \frac{1}{12} = \frac{1}{2}$$

$$P_{X|Y}(0|0) = \frac{P_{XY}(0,0)}{P(Y=0)} = \frac{1/3}{3/4} = \frac{4}{9}$$

$$P_{X|Y}(1|0) = \frac{P_{XY}(1,0)}{P(Y=0)} = \frac{5/12}{3/4} = \frac{5}{9}$$

$$P_{X|Y}(0|1) = \frac{P_{XY}(0,1)}{P(Y=1)} = \frac{1/6}{1/4} = \frac{2}{3}$$

$$P_{X|Y}(1|1) = \frac{P_{XY}(1,1)}{P(Y=1)} = \frac{1/12}{1/4} = \frac{1}{3}$$

$$P_{Y|X}(0|0) = \frac{P_{XY}(0,0)}{P(X=0)} = \frac{1/3}{1/2} = \frac{2}{3}$$

$$P_{Y|X}(1|0) = \frac{P_{XY}(0,1)}{P(X=0)} = \frac{1/6}{1/2} = \frac{1}{3}$$

$$P_{Y|X}(0|1) = \frac{P_{XY}(1,0)}{P(X=1)} = \frac{5/12}{1/2} = \frac{5}{6}$$

$$P_{Y|X}(1|1) = \frac{P_{XY}(1,1)}{P(X=1)} = \frac{1/12}{1/2} = \frac{1}{6}$$

Example: Let $X = \text{Pois}(\lambda)$ and $Y = \text{Pois}(\gamma)$ be independent.

Find the conditional distribution of X given that $X + Y = n$

$$P_{X|X+Y}(k|n) = \frac{P(X=k, X+Y=n)}{P(X+Y=n)} = \frac{P(X=k, Y=n-k)}{P(X+Y=n)}$$

$$= \frac{P(X=k) \cdot P(Y=n-k)}{P(X+Y=n)} = \frac{e^{-\lambda} \cdot \frac{\lambda^k}{k!} \cdot e^{-\gamma} \cdot \frac{\gamma^{n-k}}{(n-k)!}}{P(X+Y=n)}$$

$$\left(P(X+Y=n) = \sum_k P(X=k) \cdot P(Y=n-k) = \sum_{k=0}^n e^{-\lambda} \cdot \frac{\lambda^k}{k!} \cdot e^{-\gamma} \cdot \frac{\gamma^{n-k}}{(n-k)!} \right)$$

$$= \frac{e^{-\lambda-\gamma} \cdot \sum_{k=0}^n \frac{n!}{k!(n-k)!} \lambda^k \gamma^{n-k}}{n!} = e^{-(\lambda+\gamma)} \cdot \frac{(\lambda+\gamma)^n}{n!}$$

$$= \frac{e^{-\lambda} \cdot \frac{\lambda^k}{k!} \cdot e^{-\gamma} \cdot \frac{\gamma^{n-k}}{(n-k)!}}{e^{-(\lambda+\gamma)} \cdot \frac{(\lambda+\gamma)^n}{n!}} = \frac{n!}{k!(n-k)!} \lambda^k \gamma^{n-k} (\lambda+\gamma)^{-n}$$

$$= \binom{n}{k} \cdot \underbrace{\left(\frac{\lambda}{\lambda+\gamma} \right)^k \cdot \left(\frac{\gamma}{\lambda+\gamma} \right)^{n-k}}_{\text{Bin}(n, \frac{\lambda}{\lambda+\gamma})}$$

We can also calculate expectations under conditional probability distributions in the natural way
(whenever $P(Y = y) > 0$):

$$E[g(X)|Y = y] = \sum_x g(x)P(X = x|Y = y)$$

Example: Let X and Y be discrete random variables with

$$p_{XY}(n, m) = (1 - r)e^{-\lambda} \frac{\lambda^n r^m}{n!} \text{ for } n, m = 0, 1, \dots$$

Find $E[Y|X = Y]$

$$\begin{aligned} E[Y|X=Y] &= \sum_k k \cdot p_{Y|X=Y}(k) = \sum_{k=0}^{\infty} k \cdot \frac{P(Y=k, X=Y)}{P(X=Y)} \\ &= \sum_{k=1}^{\infty} k \cdot \frac{P(Y=k, X=k)}{P(X=Y)} = \sum_{k=1}^{\infty} k \cdot \frac{(1-r)e^{-\lambda} \frac{\lambda^k r^k}{k!}}{P(X=Y)} \end{aligned}$$

$$\begin{aligned} \left(P(X=Y) &= \sum_{n=0}^{\infty} P(X=n, Y=n) = \sum_{n=0}^{\infty} (1-r)e^{-\lambda} \frac{\lambda^n r^n}{n!} \right. \\ &= (1-r)e^{-\lambda} \sum_{n=0}^{\infty} \frac{(\lambda r)^n}{n!} = (1-r)e^{-\lambda} \cdot e^{\lambda r} = (1-r)e^{\lambda(r-1)} \left. \right) \\ &= \sum_{k=1}^{\infty} k \cdot \frac{(1-r)e^{-\lambda} \frac{\lambda^k r^k}{k!}}{(1-r)e^{\lambda(r-1)}} = \sum_{k=1}^{\infty} k \cdot e^{-\lambda r} \cdot \frac{(\lambda r)^k}{k!} \\ &= e^{-\lambda r} \cdot \sum_{k=1}^{\infty} \frac{(\lambda r)^k}{(k-1)!} \quad (j=k-1) = e^{-\lambda r} \cdot \sum_{j=0}^{\infty} \frac{(\lambda r)^{j+1}}{j!} \\ &= \lambda r e^{-\lambda r} \cdot \sum_{j=0}^{\infty} \frac{(\lambda r)^j}{j!} = \lambda r \cdot e^{-\lambda r} \cdot e^{\lambda r} = \boxed{\lambda r} \end{aligned}$$

Example: Let $X = \text{Bin}(2, p)$ and $Y = \text{Bin}(X + 1, p)$.

Find the conditional expectation of X given $Y = 0$.

$$\begin{aligned}
 P(Y=0) &= P(X=0, Y=0) + P(X=1, Y=0) + P(X=2, Y=0) \\
 &= P(Y=0|X=0) \cdot P(X=0) + P(Y=0|X=1) \cdot P(X=1) + P(Y=0|X=2) \cdot P(X=2) \\
 &= P(\text{Bin}(1, p)=0) \cdot \binom{2}{0} p^0 (1-p)^2 + P(\text{Bin}(2, p)=0) \cdot \binom{2}{1} p^1 (1-p)^1 + P(\text{Bin}(3, p)=0) \cdot \binom{2}{2} p^2 (1-p)^0 \\
 &= \binom{1}{0} p^0 (1-p)^1 \cdot (1-p)^2 + \binom{2}{0} p^0 (1-p)^2 \cdot 2p(1-p) + \binom{2}{0} p^0 (1-p)^3 \cdot p^2 \\
 &= (1-p)(1-p)^2 + (1-p)^2 \cdot 2p(1-p) + (1-p)^3 \cdot p^2 = (1-p)^3 (1+2p+p^2) = (1-p)^3 (1+p)^2
 \end{aligned}$$

$$\begin{aligned}
 E[X|Y=0] &= 0 \cdot P(X=0|Y=0) + 1 \cdot P(X=1|Y=0) + 2 \cdot P(X=2|Y=0) \\
 &= 1 \cdot \frac{P(X=1, Y=0)}{P(Y=0)} + 2 \cdot \frac{P(X=2, Y=0)}{P(Y=0)} \\
 &= \frac{1 \cdot P(X=1, Y=0) + 2 \cdot P(X=2, Y=0)}{(1-p)^3 (1+p)^2} \\
 &= \frac{1 \cdot P(Y=0|X=1) \cdot P(X=1) + 2 \cdot P(Y=0|X=2) \cdot P(X=2)}{(1-p)^3 (1+p)^2} \\
 &= \frac{P(\text{Bin}(2, p)=0) \cdot \binom{2}{1} p^1 (1-p)^1 + 2 \cdot P(\text{Bin}(3, p)=0) \cdot \binom{2}{2} p^2 (1-p)^0}{(1-p)^3 (1+p)^2} \\
 &= \frac{\binom{2}{0} p^0 (1-p)^2 \cdot 2p(1-p) + 2 \cdot \binom{2}{0} p^0 (1-p)^3 \cdot p^2}{(1-p)^3 (1+p)^2} \\
 &= \frac{2p + 4p^2}{(1+p)^2} = \frac{2p(1+p)}{(1+p)^2} = \boxed{\frac{2p}{1+p}} < 2p = E[X]
 \end{aligned}$$